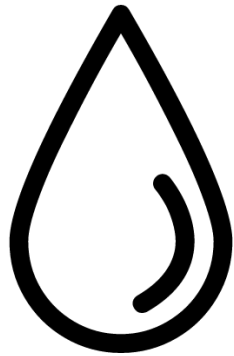


ARIZONA NATURAL RESOURCES MUSEUM

Phoenix, Arizona



Index

Executive Summary.....	5
Background.....	8
Master Plan.....	7
o Identity	
o Stewardship	
o Functionality	
Implementation.....	54
o Schedule	
o Budget	
o Staffing, Operations, Capital Costs	
Appendices.....	59
o Existing Floor Plans	
o Facilities Assessment Summary	

Executive Summary



Executive Summary

In September 2022, the University of Arizona contracted with Museum Planning, LLC to create a Master Plan to guide facility improvements and future development of 1502 West Washington Street, Phoenix, Arizona, the former Arizona Mining and Mineral Museum. The University of Arizona has an internal design/build team to manage large construction projects, and the Museum Master Plan will serve as a guide for the museum design/build process. Mark Walhimer of Museum Planning, LLC worked closely with University of Arizona Associate Vice President of Research Institutes Dr. Elliott Cheu, the project sponsor; Catherine Carter Sandoval, the Museum Curator; and the larger University of Arizona staff in a planning process that consisted of a detailed assessment of University of Arizona programs and facilities, an articulation of community needs and institutional objectives, and the generation of plans to guide the future development of the museum. The proposed name for the new institution is the Arizona Natural Resources Museum.

Process

Assessment and planning were guided by an approach emphasizing financial sustainability, low capital costs, and rapid implementation. Instead of looking back, the approach was focused on moving the previous Arizona Mining and Mineral Museum toward a new future under the leadership of the University of Arizona. The process did not emphasize things that were lacking; rather, it identified opportunities for advancing the Arizona Natural Resources Museum in alignment with the communities it serves. This work resulted from engagement and collaboration with the University of Arizona, area museums, State Representative Gail Griffin, and museum staff and community members through in-person and virtual meetings and workshops.

Key findings from this process include:

- The creation of a new interactive museum focused on Arizona natural resources is feasible and met with museum community support, local community support, and business leader support.
- Through improved marketing, new strategic partnerships, local and regional collaborations, and inclusive programming, the Arizona Natural Resources Museum will reach and impact local and tourist audiences.
- With sustained and consistent community outreach programs, the reopened museum will reflect and better serve a diverse local community.
- There are revenue opportunities for facility rentals for legislator events, university events, and on-site museum fundraising events and gatherings.
- Used as a financial benchmark, the Tucson-located (two-hour drive) Flandrau Science Center and Planetarium attendance and finances support the case for a roughly equivalent museum in Phoenix.
- Short-range planning and a museum accreditation (ASTC, AAM) process will provide the Arizona Natural Resources Museum with guidance and opportunities to assess program quality, develop a strong collections policy, streamline bureaucratic procedures, and improve hiring and staff development.
- The Arizona Natural Resources Museum can leverage its historic building and prime downtown locations, which outwardly reflect and engage the public on-site and throughout downtown and adjacent neighborhoods.

Although not initially identified as part of the planning process, it became clear that there was a need for a better vehicle to articulate the mission, vision, and values of the Arizona Natural Resources Museum. A museum website was created to foster as transparent and inclusive a process as possible.

Executive Summary

Master Plan

This Master Plan is informed by the findings of our assessments and in alignment with the vision and objectives articulated in Arizona State Bill 1415 (introduced by Senator Griffin, Representative Barton, and Representative Bowers); the Arizona Mining, Mineral & Natural Resource Education Museum Implementation Plan (2016); Mineral, Mining Artifacts, and Physical Assets of the former Mining and Mineral Museum (2017); and the University of Arizona College of Science Team Report (October 2016). We have created an actionable, flexible, and prioritized plan for recommended improvements to the museum building, facilities, and operations. Recommendations are centered on strengthening the museum's identity and presence, guiding the stewardship of its collections and facilities, and prioritizing functional improvements to enhance program capacity.

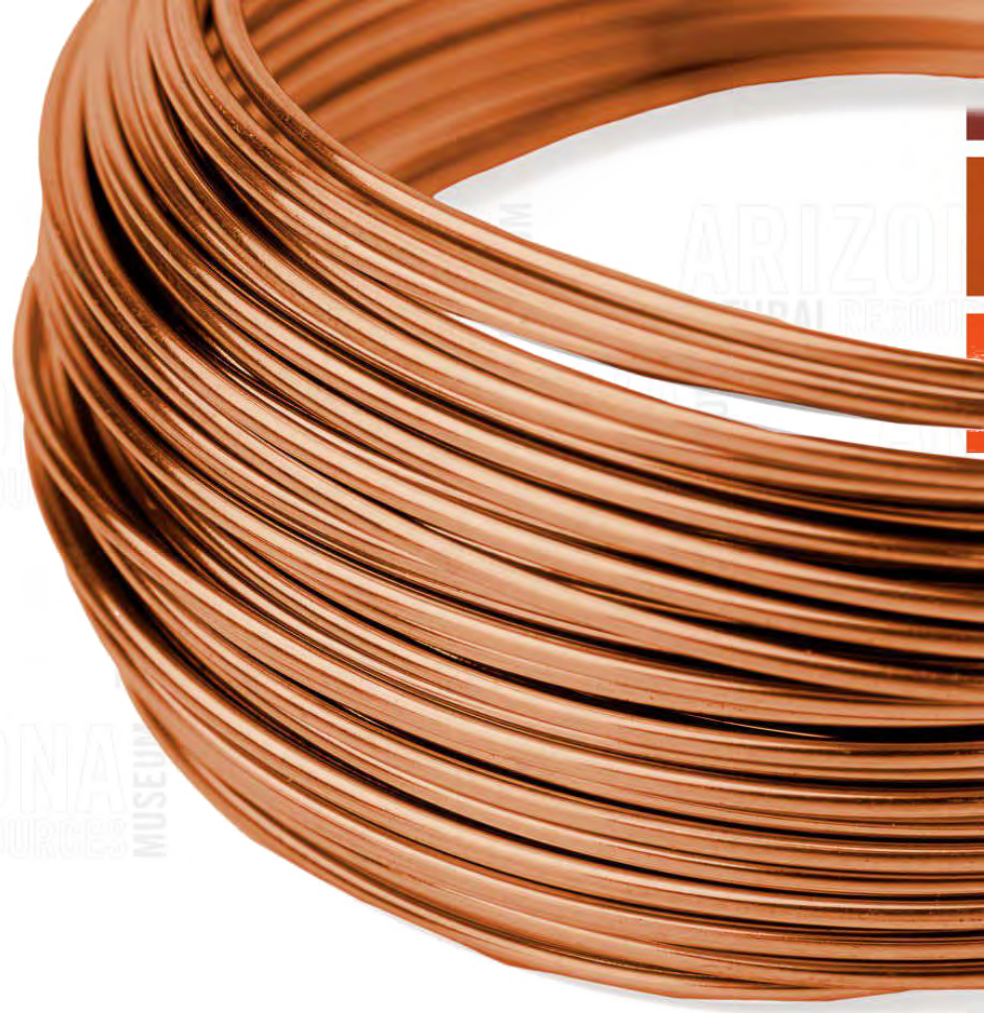
Principal recommendations include:

- Establish a strong, singular Arizona Natural Resources Museum identity, anchored at 1502 West Washington Street and focused on Arizona's interrelated science, art, and history.
- Reflect community voices and identities in the outward appearance of the museum, its urban footprint, and the eventual growth of programs through local and regional partnerships.
- Address needed repairs and building code concerns and create museum-specific plans to improve maintenance and operations, better coordinate improvements with preservation objectives, and guide environmental and financial sustainability objectives.
- Pursue opportunities on-site and within the existing museum footprints to expand event spaces, better distribute the mix of permanent and temporary galleries, and connect learning spaces with exhibits and the larger urban context.
- Create a single point of arrival and access to the museums, resolve visitor and staff circulation routes, and consolidate back-of-house functions.

Implementation

Proposals offered in this Master Plan are intended to provide a menu of prioritized options for achieving the Arizona Natural Resources Museum goals. They are to be pursued in continued consultation with strategies of the University of Arizona and museum stakeholders, and with adequate funding from the University of Arizona and the State of Arizona.

Master Plan



Master Plan

■ Do you know where your consumer items come from?

They are either grown or mined.

Introduction

The Arizona Natural Resources Museum (ANRM) (working title) will be a highly interactive museum designed to attract and educate K-12 students, school groups, educators, college students, and family audiences. The museum will be an immersive experience centered on Arizona's natural resources education.

Primary natural resources to be covered in the ANRM:

1. Minerals
2. Water
3. Food
4. Forests

Mission, Vision, and Values

Mission:

To serve Phoenix and the larger Arizona community by showcasing the importance of natural resources, including minerals, water, forests, and food, and the innovations that will help provide these resources for future generations.

Vision:

To become the top destination in Arizona focusing on the state's abundant natural resources.

Values:

An organization that fuels innovation, ignites minds, and provides appreciation and wonder for the natural world.

Content Areas: (water a thread throughout each content area)

1. Water, Minerals, Arizona Nature, Forests, and Food

Introduction

We all require natural resources. The ANRM will personalize our needs for natural resources through thought-provoking interactive exhibits on Arizona water, minerals, forests, mining, and food. The museum will showcase research at the University of Arizona that impacts and fosters these natural resources.

The exhibitions will engage present and future STEM students, nurturing and inspiring the next generation of scientists and engineers. The visitor experience will include tactile, open-ended, and didactic interactions, and will constantly change, with monthly updates to exhibits, lectures, programming, and museum outreach.

Each of us requires natural resources to live; however, natural resources require creative solutions to ensure a sustainable future. The ANRM will be a compelling space for engagement, focused on the extraordinary natural resources of the state of Arizona and the innovations and research of the University of Arizona.

Museum Programming

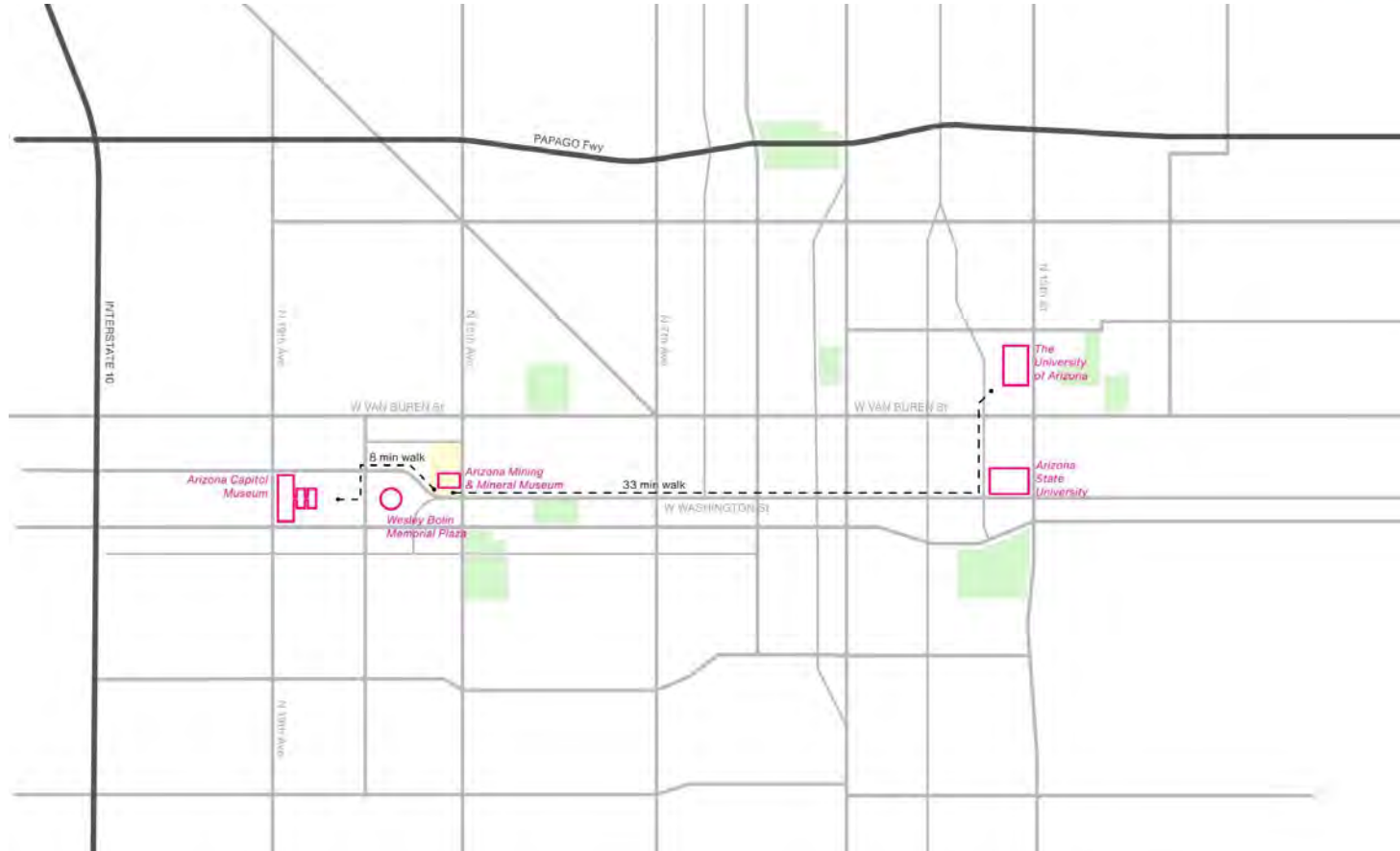
Museum programming will be updated monthly with new lectures, community outreach, and individual exhibits (one exhibit rather than an exhibition).

Exhibit areas will change once or twice per year. For example, the mining area might be reconfigured once a year.

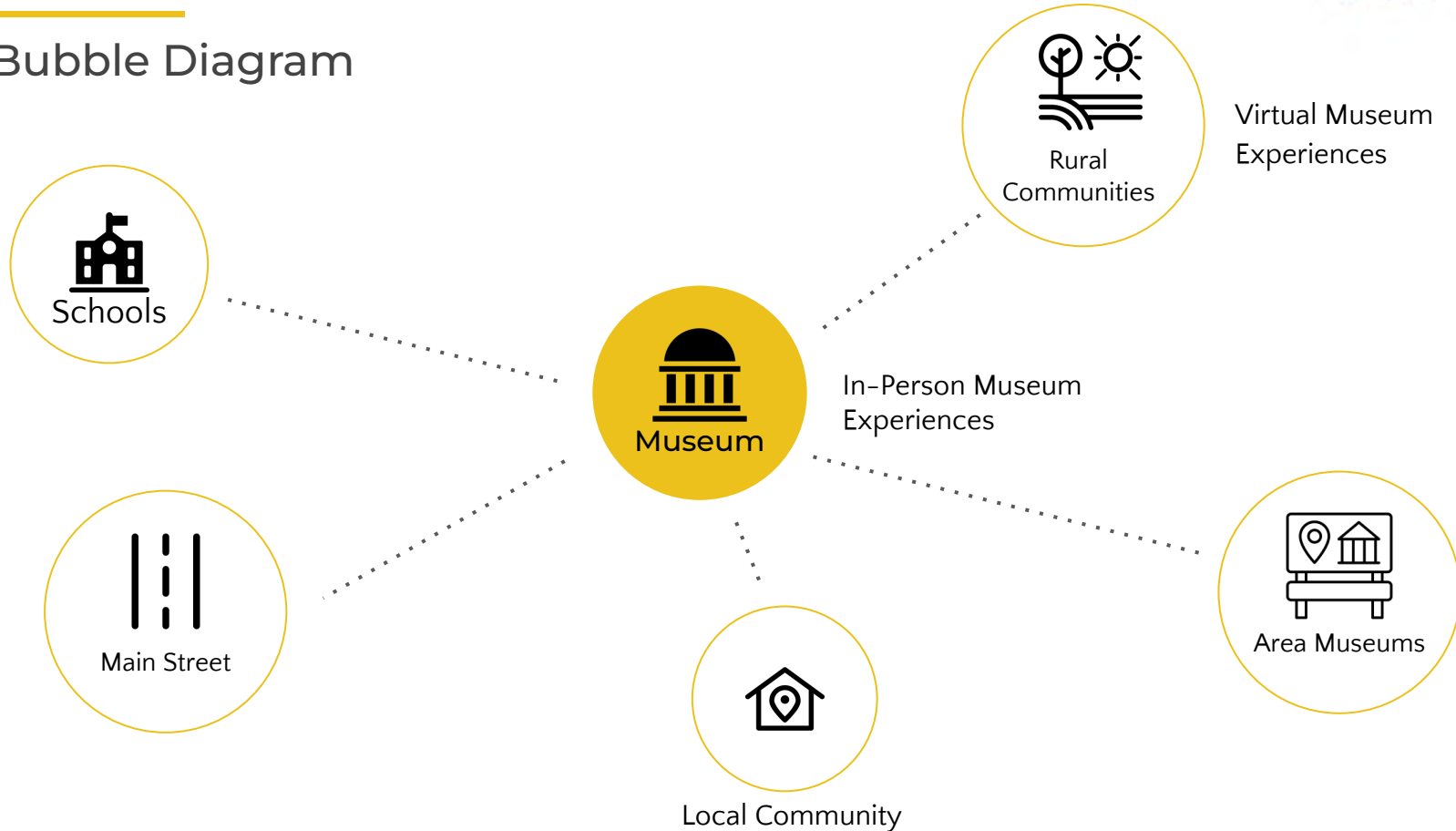
The current size of the changing gallery is manageable. For example, at the current size, it should be possible to create a new exhibition every month (remotely updateable digital signage, graphics, and one or two manipulatives).

Continually updating museum programming is vital to success; it is the only way to drive repeat visitation. That is an advantage to a university museum in utilizing the research and involvement of students, staff, and volunteers, and creating a museum community.

- Exhibit gallery updated monthly
- New museum programming monthly—new lectures, community outreach, and individual exhibits
- Change each exhibit area—Food, Forest, Arizona Nature, Water, and Mining—on a rotating basis every six to eight months



Bubble Diagram



Exterior



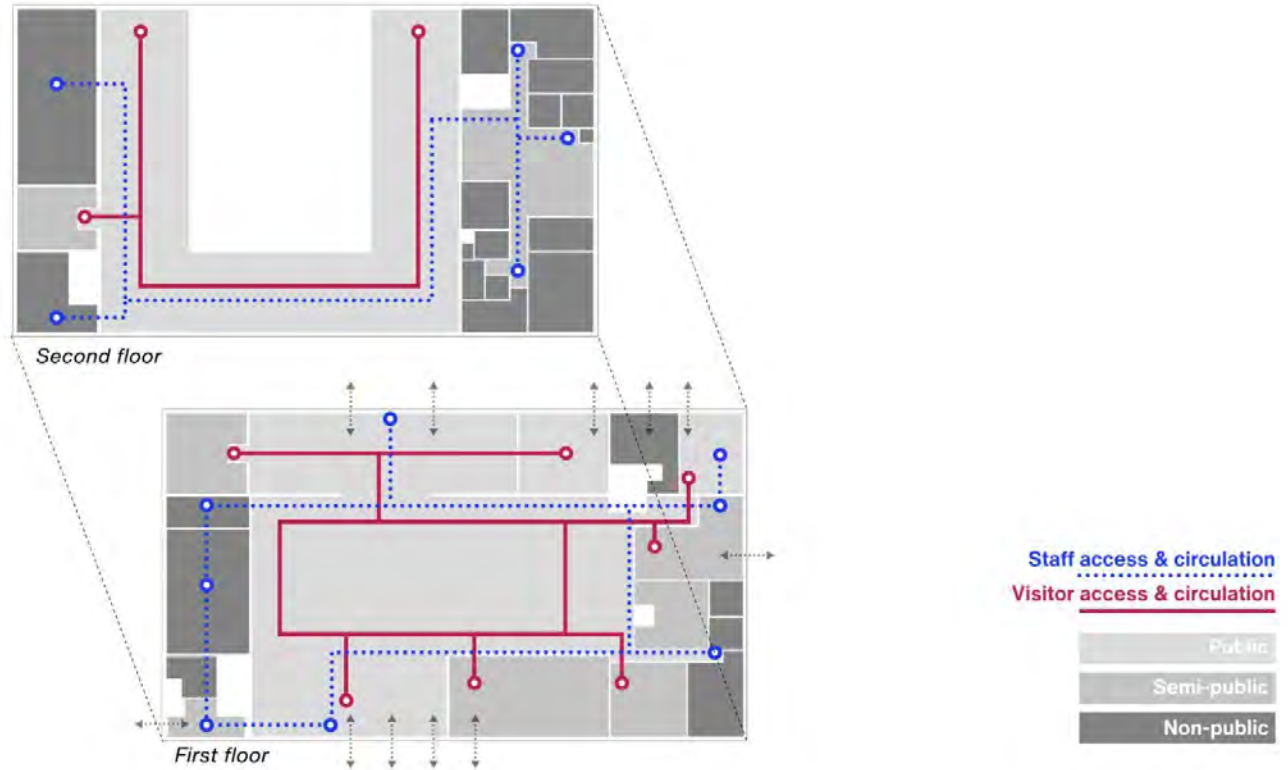
Exterior



Exterior - Museum Entrance



Circulation



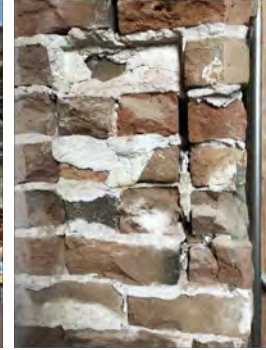
Exterior



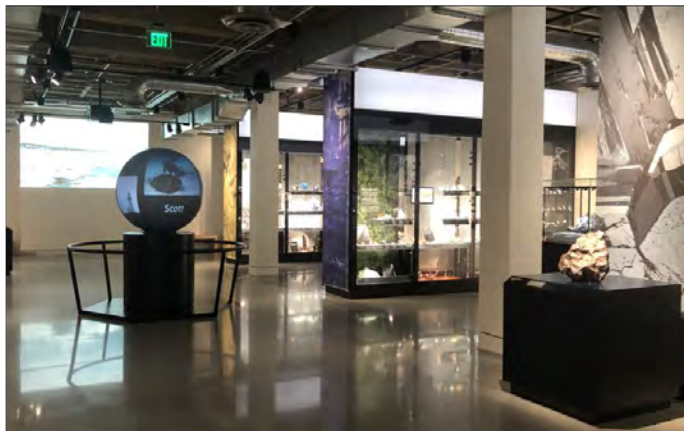
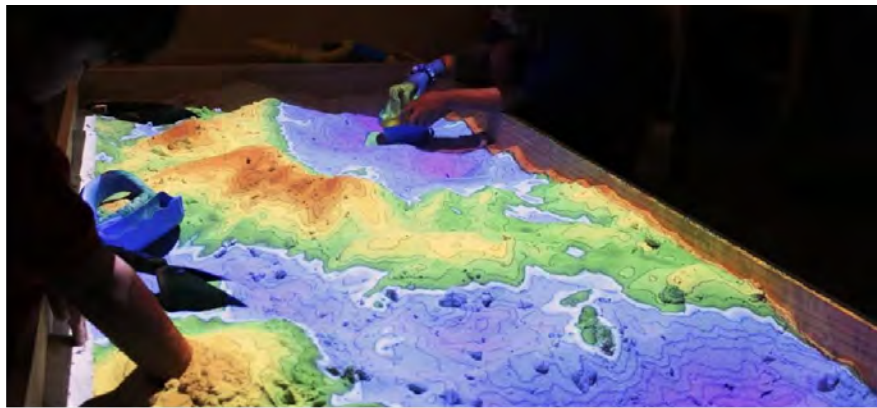
Exterior



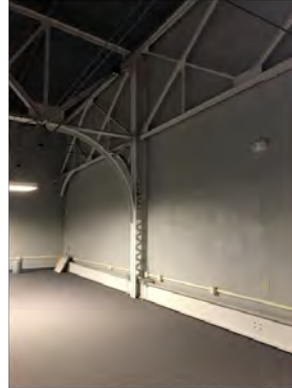
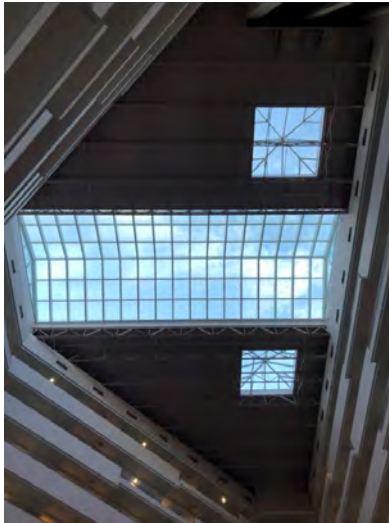
First Floor



First Floor



Second Floor



Second Floor

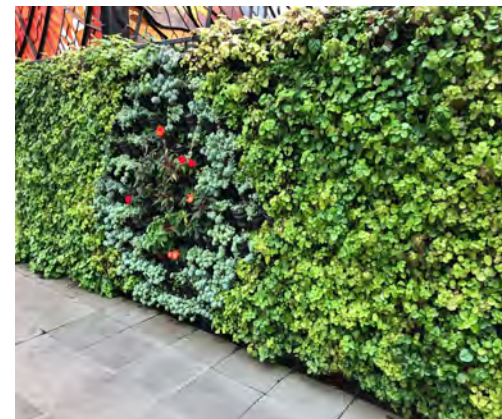


Exhibit Areas

First Floor

- Arizona Water
- Mining
- Arizona Mineral Gallery
- Overhead Earth
- Open Storage
- Multipurpose Room
- Classroom 1
- Classroom 2

Second Floor

- Changing Gallery
- Video Wall
- Arizona Nature
- Forests
- Food
- Makerspace

Exterior

- Science Playground
- Xeriscaping
- Projection-Mapping
- LED Digital Sign of Events
- Community Signage
- Signage to Capital Museum
- Signage to the University of Arizona Medical Campus

Exhibit Areas

Outreach

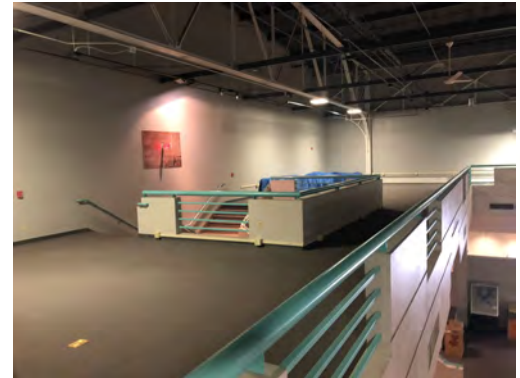
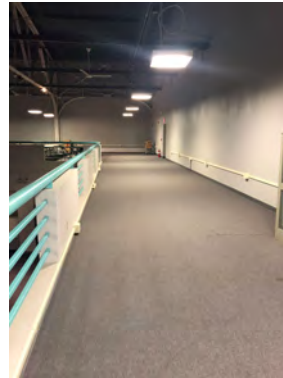
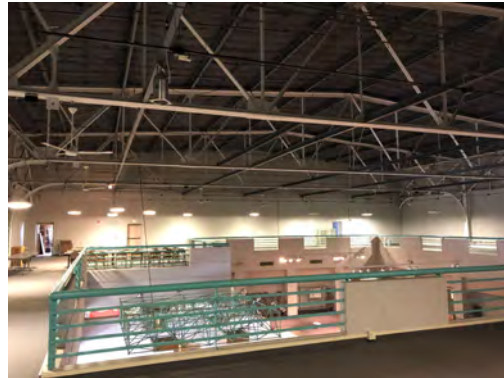
- Virtual Museum
- Community Outreach
- Museum Van

Online

- Website: <https://arizona-resources.org>
Password: nature
- Museum Visit Information
- Museum Collection
- Virtual Museum

First Floor





First Floor

1. **Lobby**—Entrance desk, ticket sales.
2. **Museum Store** and stockroom.
3. **Your Innovation Badge**—Your journey to becoming the next groundbreaking innovator starts with your own curiosity and creativity. By answering a few questions, your journey becomes personalized to your interests and tracks your innovation at the museum. It also provides you with the opportunity to continue your work at home.
4. **Introduction**—An overview of the Arizona Natural Resources Museum experience.
5. **Water**—This changing gallery will showcase the groundbreaking innovations of the University of Arizona as inspiration to create your own solutions to today's water problems.
6. **Arizona Mining**—The Copper State is the location of extraordinary natural wonders. This gallery will highlight the challenges and opportunities in agriculture, mining, energy production, and climate change.
7. **Tree**—A central tree will serve as an icon for the visitor experience.
8. **Mineral Gallery**—With one of the world's finest collections of minerals (more than 20,000 specimens), this interactive gallery will use virtual reality and augmented reality to explain the science and suggest future uses.
9. **Open Storage and Laboratory**—The former stage area is now a glass-enclosed specimen storage space and laboratory viewable from the first floor.
10. **Multipurpose Room**—A room that serves as a theater, lecture space, and space for special events.
11. **Earth and Innovation Scoreboard**—The overhead Earth spherical projection also serves as an “innovation scoreboard” that displays the current innovation projects created by visitors to the museum.
12. **School Innovate**—School classes (and summer camps) can sign up for guided school innovation programs.
13. **Special Events**—First floor can be rented for special events (dinners, cocktail parties, birthday parties, and meetings).
14. **2 School Group Classrooms**
15. **Exhibit Support**—Exhibition shop and staff conference room.
16. **Catering Kitchen**

First Floor



First Floor _ Entry



First Floor _ Water



First Floor _ Water + Mining Area



First Floor _ Mining Area



First Floor _ Minerals Area



First Floor _ Multipurpose Room



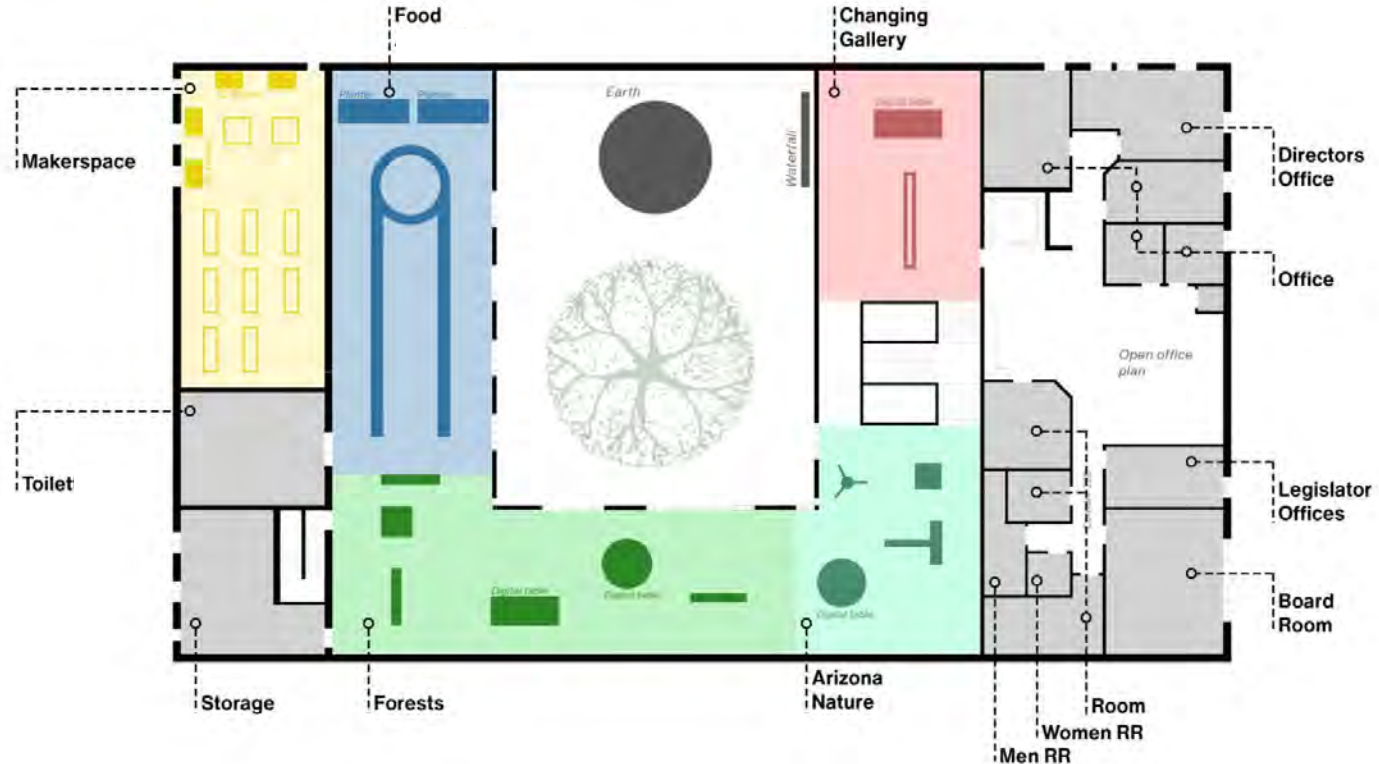
First Floor _ Multipurpose Room



Second Floor

1. **Changing Gallery**—Entrance desk, ticket sales.
2. **Video Wall**
3. **Arizona Nature**—A natural resources gallery showcasing trees, grasses, animals, and livestock of Arizona.
4. **Forests**—A gallery showcasing this important natural resource.
5. **Food**—Automation, artificial intelligence, and machine learning will be vital to the future of Arizona's food and agriculture.
6. **Makerspace**—A continuation of Food and Agriculture, where visitors can 3D print, solder, and learn basic robotics and computer code.
7. **Natural Resources Sculpture**—A ceiling-mounted kinetic sculpture that is part of an twice-daily Arizona natural resources object theater.
8. **Board Room**
9. **Staff Offices and "Hot Desks"**
10. **Legislative Offices**

Second Floor



Second Floor _ Arizona Nature



Second Floor _ Arizona Nature



Second Floor _ Forests



Second Floor _ Food



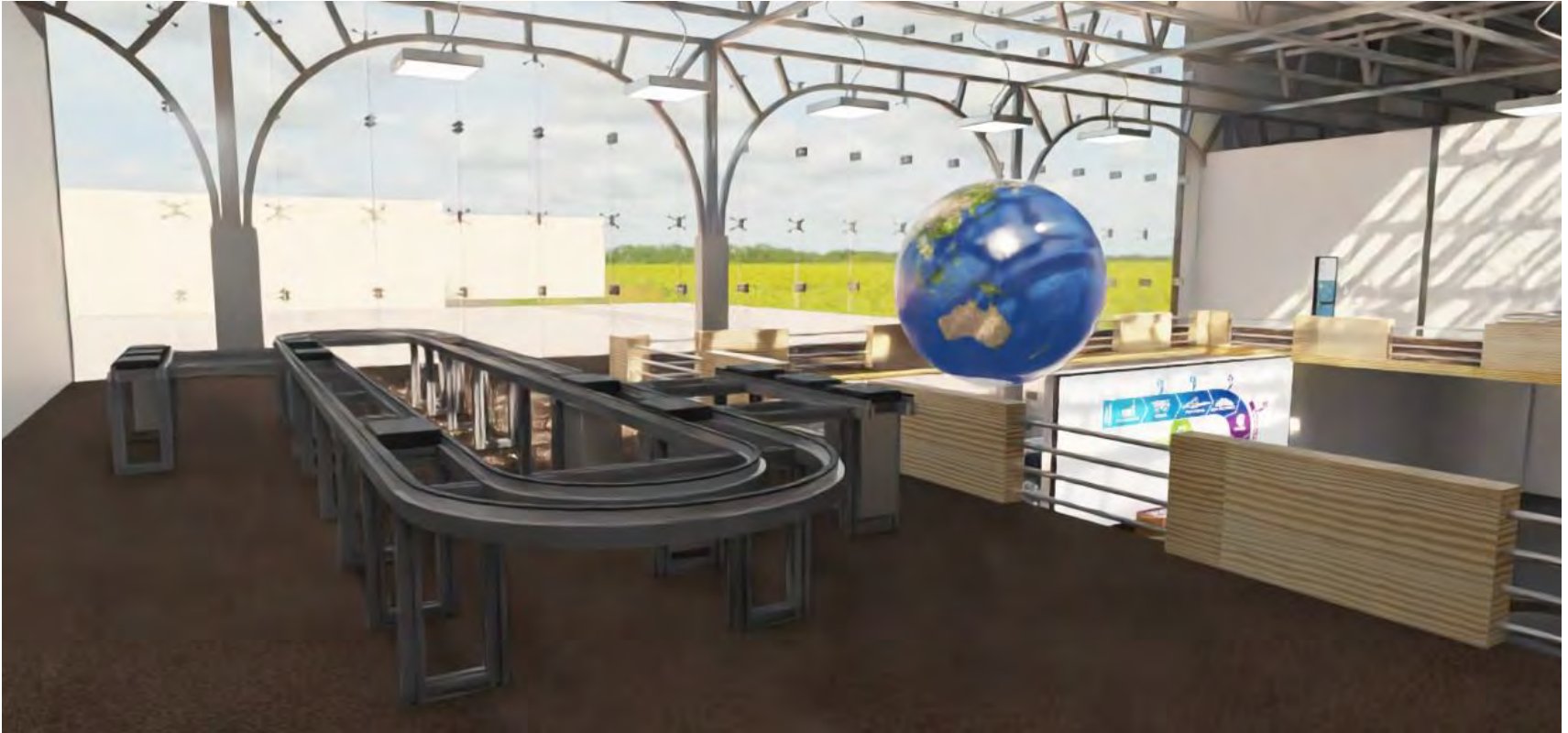
Second Floor _ Food



Second Floor _ Food



Second Floor _ Food



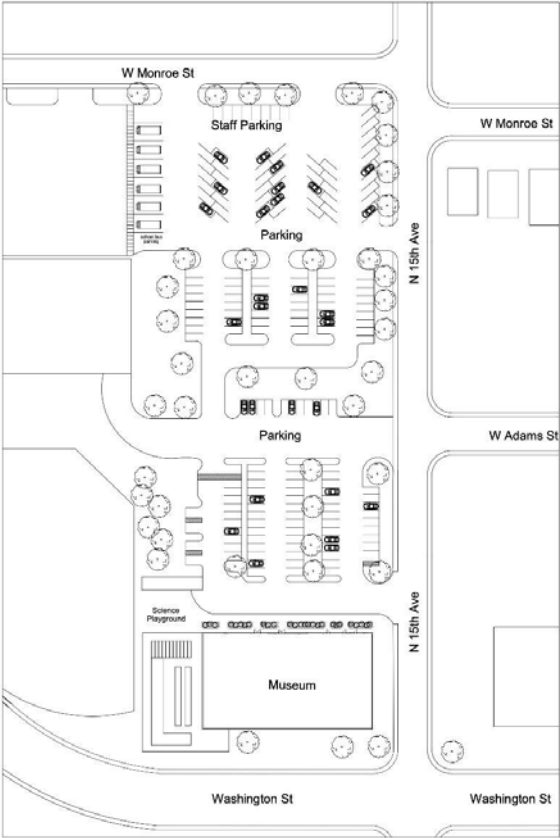
Second Floor _ Changing Gallery



Exterior

- 1. Science playground
- 2. Xeriscaping at the University of Arizona
- 3. Projection-mapping
- 4. LED digital sign of events
- 5. Monument signs with LED text panels
- 6. Parking lot exhibits
- 7. Community signage
- 8. Signage to Capital Museum
- 9. Signage to the University of Arizona Medical Campus
- 10. Outdoor demonstration space w/ MeeFog

Exterior



3 Building Plan: Site Plan
Not to scale

Logo & Style Guide



ARIZONA
NATURAL RESOURCES MUSEUM

The logo features the word "ARIZONA" in a large, bold, dark red sans-serif font. Below it, the words "NATURAL RESOURCES" are in a smaller, bold, sans-serif font, with "NATURAL" in light blue and "RESOURCES" in orange. To the right of "RESOURCES", the word "MUSEUM" is written vertically in a bold, orange sans-serif font.

Logo & Style Guide

TRADE GOTHIC LT STD
CONDENSED NO. 18
CONDENSED NO. 18 OBLIQUE
BOLD CONDENSED NO. 20
BOLD CONDENSED NO. 20 OBLIQUE
REGULAR | *OBLIQUE*
BOLD | *BOLD OBLIQUE*

TYPEFACE

Logo & Style Guide



Logo & Style Guide



Implementation



Implementation



Budget

Schedule

Staffing Plan

Operations

Building Schedule

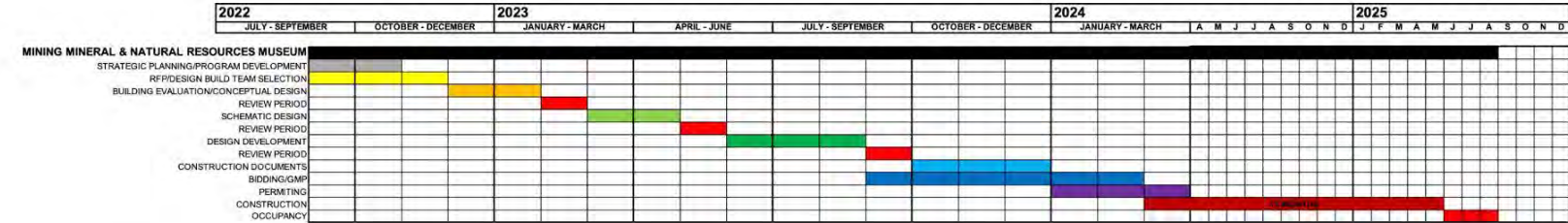


THE UNIVERSITY OF ARIZONA
BUSINESS AFFAIRS
Planning, Design
& Construction

MINING MINERAL & NATURAL RESOURCES MUSEUM

DRAFT - 07/28/2022

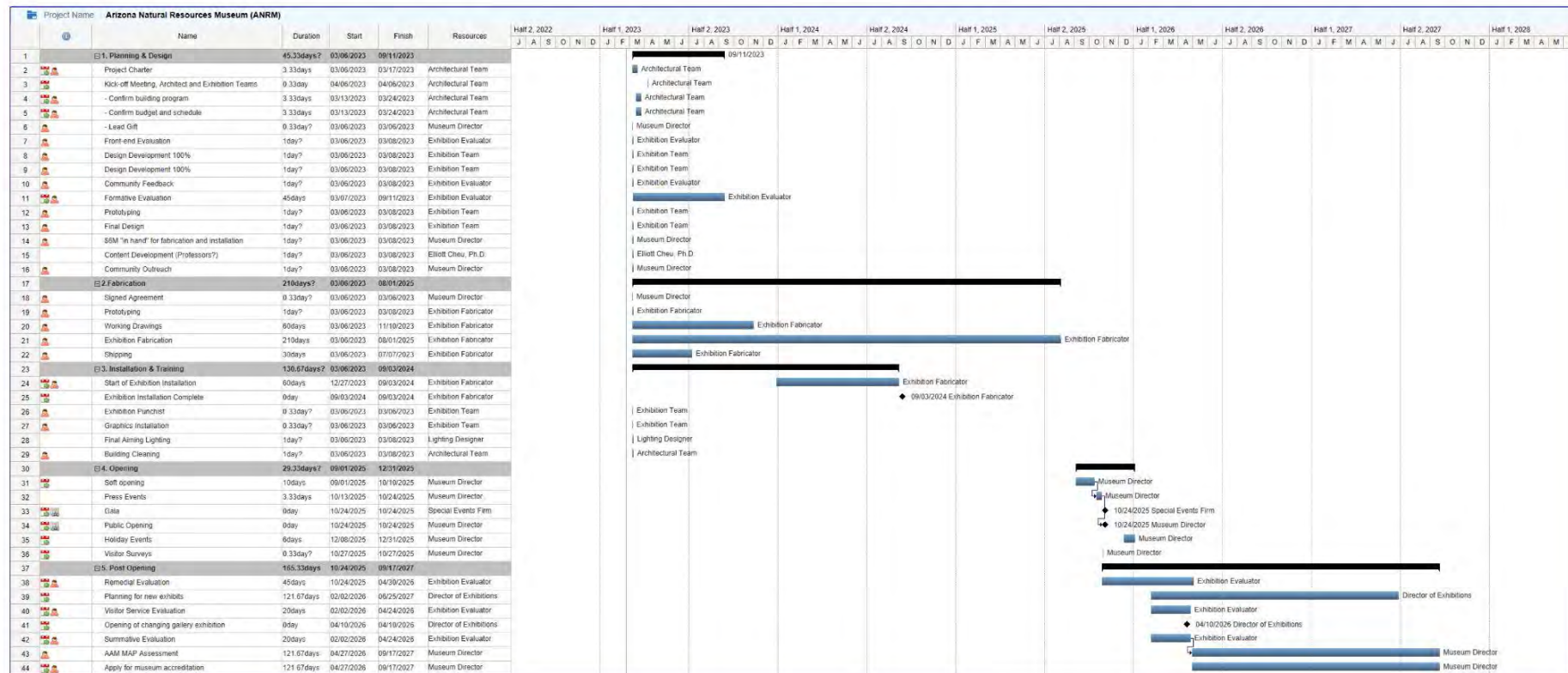
SCHEDULE



- STRATEGIC PLANNING/PROGRAM DEVELOPMENT
- RFP/DESIGN BUILD TEAM SELECTION
- BUILDING EVALUATION/CONCEPTUAL DESIGN
- REVIEW PERIOD
- SCHEMATIC DESIGN
- REVIEW PERIOD
- DESIGN DEVELOPMENT
- REVIEW PERIOD
- CONSTRUCTION DOCUMENTS
- BIDDING/GMP
- PERMITTING
- CONSTRUCTION
- OCCUPANCY



Exhibition Schedule



Benchmarks

- [Flandrau Science Center and Planetarium, Tucson, Arizona](#)
- [Biosphere 2, Oracle, Arizona](#)
- [Outdoor Adventure Center \(OAC\), Detroit, Michigan](#)
- [Children's Museum Tucson, Tucson, Arizona](#)
- [Children's Museum of Phoenix, Phoenix, Arizona](#)
- [Arizona Science Center, Phoenix, Arizona](#)
- [The Discovery, Reno, Nevada](#)

Appendices



Appendices

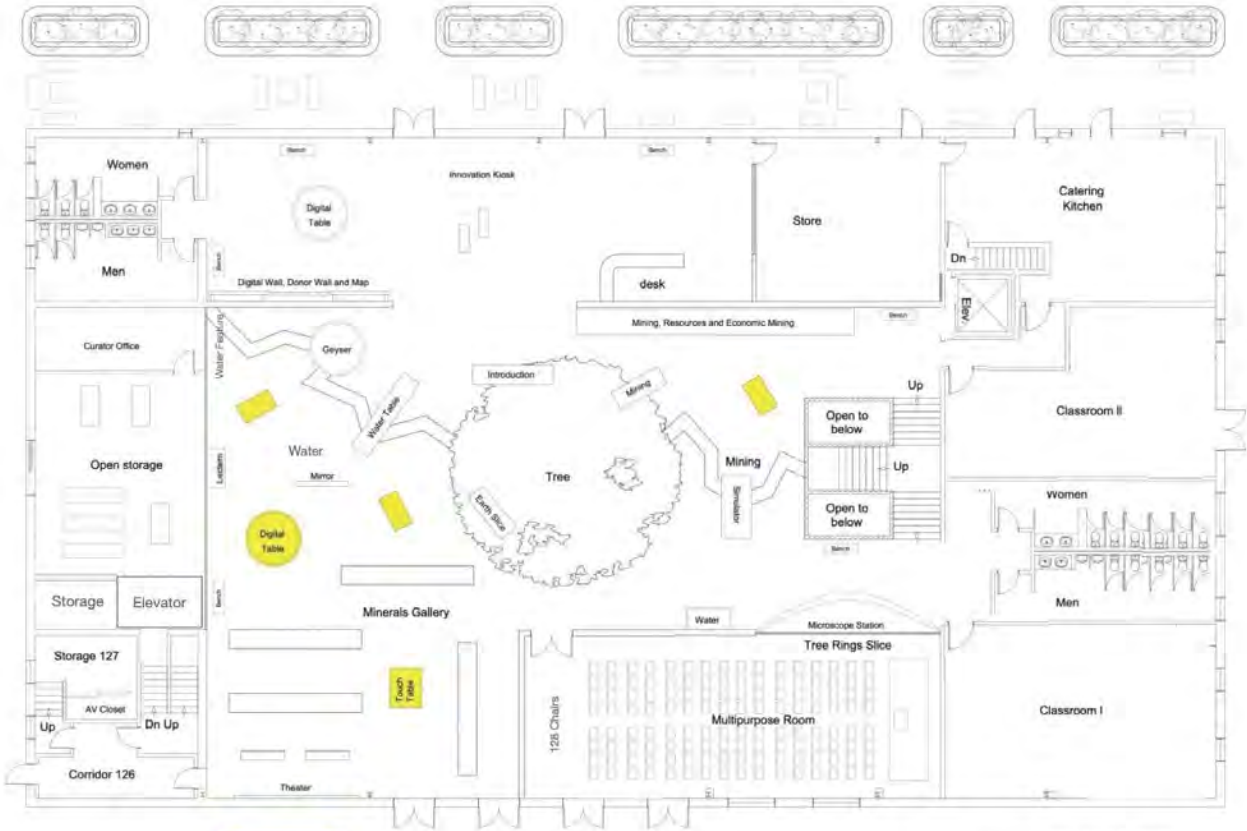
Floor Plans

Survey Summaries

Meeting Notes

First Floor

Science Playground



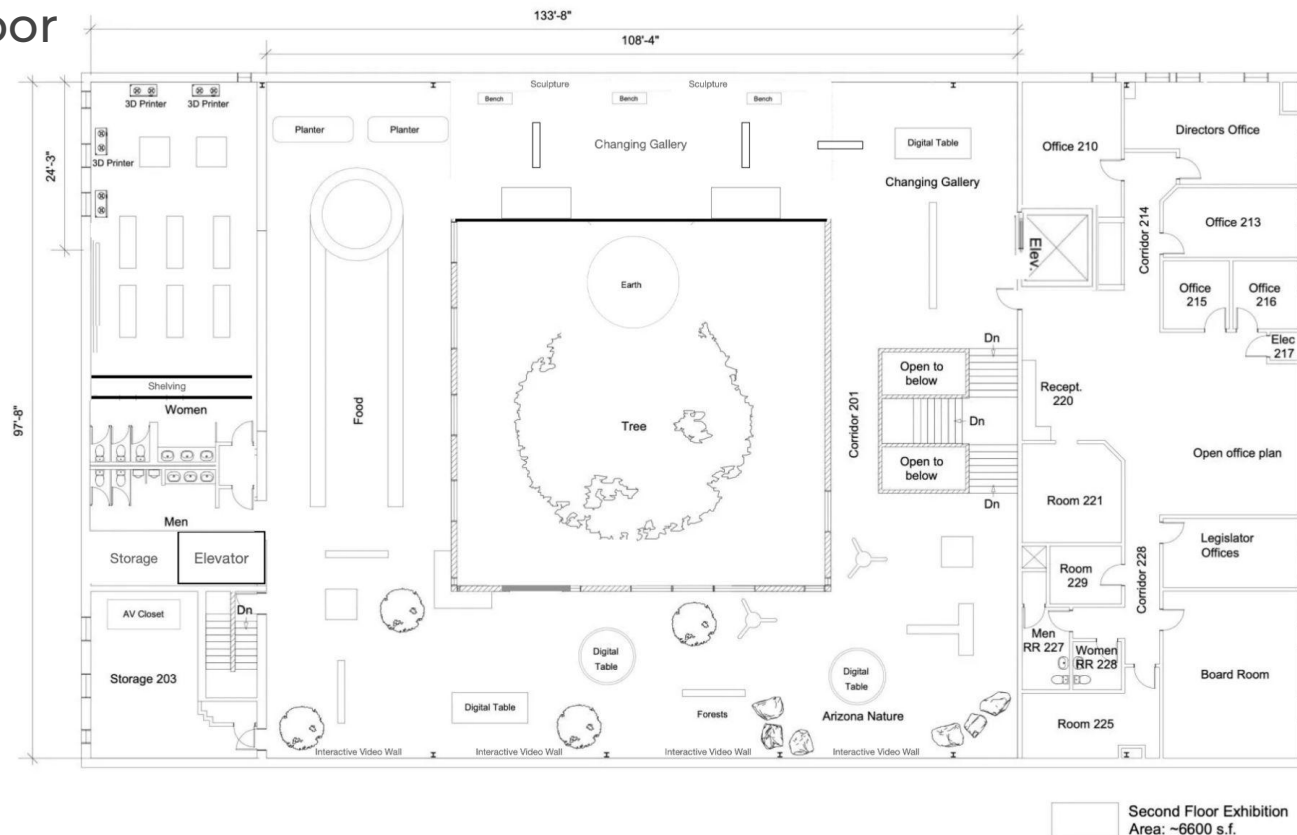
1 Building Plan: First Floor
Scale: 1/16" = 1'-0"

Movable Exhibits

Greenscaping

First Floor Exhibition Area: ~8200 s.f.

Second Floor



1

Building Plan: Second Floor

Scale: 1/16" = 1'-0"

Basement



1

Building Plan: Basement

Scale: 1/16" = 1'-0"







Appendices

<https://arizona-resources.org/about> password: nature

Presentation link: <https://bit.ly/AZ-Resources>

ARIZONA NATURAL RESOURCES MUSEUM

Phoenix, Arizona